

Math 208 Conceptual Problems #6 (Sections 6.1-6.2) - SOLUTIONS

- (1) Google the word ‘eigenfaces’ and look at the Wikipedia page, which has some pictures. (They come from artificial intelligence research on computer vision).

Here, a vector $\mathbf{v}$ represents an image. Basically $\mathbf{v}$ is the list of RGB color values of each pixel in the image, so $\mathbf{v} \in \mathbb{R}^N$ for some very large N . An ‘eigenface’ is an eigenvector for a matrix related to ‘image vectors’.

(This is not a homework problem – just a neat application of linear algebra that’s outside the scope of Math 308.)

- (2) (Practice showing something is a subspace) Suppose λ is an eigenvalue for the matrix A . Consider the λ -eigenspace of A :

$$E_\lambda(A) = \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \lambda\mathbf{v}\},$$

the set of all vectors $\mathbf{v}$ satisfying the equation $A\mathbf{v} = \lambda\mathbf{v}$. One reason why $E_\lambda(A)$ is a subspace is because it is the nullspace of $A - \lambda I$. Show that $E_\lambda(A)$ is a subspace by directly checking the three conditions needed to be a subspace.

Solution. First, we have to check that $\mathbf{0} \in E_\lambda(A)$: this is true because $A\mathbf{0} = \mathbf{0} = \lambda\mathbf{0}$ (regardless of what the eigenvalue λ is).

Next, suppose $\mathbf{v} \in E_\lambda(A)$ and $\mathbf{u} \in E_\lambda(A)$. This means that $\mathbf{v}, \mathbf{u}$ are vectors such that $A\mathbf{v} = \lambda\mathbf{v}$ and $A\mathbf{u} = \lambda\mathbf{u}$. Then

$$A(\mathbf{v} + \mathbf{u}) = A\mathbf{v} + A\mathbf{u} = \lambda\mathbf{v} + \lambda\mathbf{u} = \lambda(\mathbf{v} + \mathbf{u}).$$

Since $A(\mathbf{v} + \mathbf{u}) = \lambda(\mathbf{v} + \mathbf{u})$, this means $\mathbf{v} + \mathbf{u} \in E_\lambda(A)$.

Finally, suppose $\mathbf{v} \in E_\lambda(A)$ and $r \in \mathbb{R}$ is a scalar. So we know $A\mathbf{v} = \lambda\mathbf{v}$. Then

$$A(r\mathbf{v}) = r(A\mathbf{v}) = r(\lambda\mathbf{v}) = \lambda(r\mathbf{v}).$$

Since $A(r\mathbf{v}) = \lambda(r\mathbf{v})$, this means that $r\mathbf{v} \in E_\lambda(A)$. Therefore $E_\lambda(A)$ is a subspace.

(3) Let $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$.

- (a) Compute the eigenvalues and eigenvectors of A .

Solution. The eigenvalues of A are the roots of the polynomial $\det(A - \lambda I) = 0$.

$$\det \begin{bmatrix} 4 - \lambda & 1 \\ 3 & 2 - \lambda \end{bmatrix} = \lambda^2 - 6\lambda + 5 = (\lambda - 5)(\lambda - 1) = 0.$$

Therefore, the eigenvalues are 1 and 5.

To compute the eigenvectors of $\lambda = 1$, we need to find a basis for the nullspace of $A - I = \begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix}$. The eigenvector $\lambda = 1$ has eigenspace spanned by

$\mathbf{w} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. Similarly, check that the eigenvalue $\lambda = 5$ has eigenspace spanned by the eigenvector $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- (b) If λ is an eigenvalue of A then there is a vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. Using this equation, show that λ^2 is an eigenvalue of A^2 . What is an eigenvalue of A^{-1} ? Now compute all eigenvalues and eigenvectors of A^2 and A^{-1} .

Solution. $A\mathbf{v} = \lambda\mathbf{v} \Rightarrow A^2\mathbf{v} = \lambda A\mathbf{v} = \lambda^2\mathbf{v}$. Therefore, λ^2 is an eigenvalue of A^2 with eigenvector $\mathbf{v}$. Similarly, $A\mathbf{v} = \lambda\mathbf{v} \Rightarrow A^{-1}A\mathbf{v} = \lambda A^{-1}\mathbf{v}$ which means that $A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$. Therefore, $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} with eigenvector $\mathbf{v}$. Using this information we see that 1 and 25 are eigenvalues of A^2 . Since A^2 is a 2×2 matrix it has no further eigenvalues. The eigenvectors for each eigenvalue are the same as for A . Similarly, the eigenvalues of A^{-1} are 1 and $\frac{1}{5}$ with same eigenvectors as A .

- (c) Find a matrix B that shares an eigenvector with A but has different eigenvalues.

Solution. Suppose we try to find $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Then we need

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a+b \\ c+d \end{bmatrix} = \mu \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

This means that we need $a+b = \mu$ and $c+d = \mu$. There are infinitely many solutions, but we could for instance take $\mu = 3$ and $B = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$. Then A and

B share the eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ but have different eigenvalues. Check that B has only 3 as eigenvalue and the eigenspace of 3 is all of $\mathbb{R}^2$.

- (d) Find an invertible matrix P and a diagonal matrix D so that $A = PDP^{-1}$. Then, compute A^{1000} .

Solution. To diagonalize A , we use the eigenvectors and values we found above. We'll get

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}^{-1}.$$

Note that $\begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}.$

$A^k = PDP^{-1}PDP^{-1} \dots PDP^{-1}$ where the right hand side is the product of PDP^{-1} with itself repeated k times. Therefore, $A^k = PD^kP^{-1}$. In our case,

$$A^k = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 5^k & 0 \\ 0 & 1 \end{bmatrix} \cdot \frac{1}{4} \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 + 3(5^k) & -1 + 5^k \\ -3 + 3(5^k) & 3 + 5^k \end{bmatrix}.$$

This is valid for any k , including $k = 1000$.

- (e) Suppose $\mathbf{v}$ is an eigenvector of an arbitrary matrix M , with eigenvalue λ . Show (using matrix algebra) that $\mathbf{v}$ is also an eigenvector of $M + I$, but with a different eigenvalue. What eigenvalue is it?

Solution. Proceeding algebraically: $(M + I)\mathbf{v} = M\mathbf{v} + I\mathbf{v} = \lambda\mathbf{v} + \mathbf{v} = (\lambda + 1)\mathbf{v}$. So, $\mathbf{v}$ is an eigenvector of $M + I$ with eigenvalue $\lambda + 1$.

(4) (Reflections and projections)

(a) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation from the conceptual problems for Chapter 4:

$$T(\mathbf{x}) = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ -2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} \mathbf{x}.$$

Determine the eigenvalues of T , and find a basis for each eigenspace.

Solution. Be careful to keep track of the $\frac{1}{3}$ factor when computing $A - \lambda I$!

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} \frac{-1}{3} - \lambda & \frac{-2}{3} & \frac{2}{3} \\ \frac{-2}{3} & \frac{2}{3} - \lambda & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} - \lambda \end{bmatrix} \\ &= \frac{1}{27} \det \begin{bmatrix} -1 - 3\lambda & -2 & 2 \\ -2 & 2 - 3\lambda & 1 \\ 2 & 1 & 2 - 3\lambda \end{bmatrix} \quad (\text{factoring out } \frac{1}{3} \text{ from each row}). \\ &= \frac{1}{27} \det \begin{bmatrix} -1 - 3\lambda & -2 & 2 \\ -2 & 2 - 3\lambda & 1 \\ 0 & 3 - 3\lambda & 3 - 3\lambda \end{bmatrix} \quad (\text{applying } R_2 + R_3 \rightarrow R_3) \\ &= \frac{1}{27} (3 - 3\lambda) \det \begin{bmatrix} -1 - 3\lambda & -2 & 2 \\ -2 & 2 - 3\lambda & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad (\text{factoring out } 3 - 3\lambda \text{ from the third row}) \\ &= \frac{1}{27} (3 - 3\lambda) \det \begin{bmatrix} -1 - 3\lambda & -4 & 0 \\ -2 & 1 - 3\lambda & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad (\text{applying } R_2 - R_3 \rightarrow R_2 \text{ and } R_1 - 2R_3 \rightarrow R_1) \\ &= \frac{1}{27} (3 - 3\lambda) \det \begin{bmatrix} -1 - 3\lambda & -4 \\ -2 & 1 - 3\lambda \end{bmatrix} \quad (\text{cofactor expansion along third column}) \\ &= \frac{1}{27} (3 - 3\lambda) (9\lambda^2 - 9). \end{aligned}$$

Whew! That simplifies to $0 = -(\lambda - 1)^2(\lambda + 1)$, so the solutions are $\lambda = 1$ with multiplicity two, and $\lambda = -1$ with multiplicity one.

To find the eigenspaces, we set $\lambda = 1$ or -1 , and compute $\text{null}(A - \lambda I)$. This computation is more straightforward, and the eigenspaces are:

$$\lambda = 1 : \text{span} \left\{ \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \lambda = -1 : \text{span} \left\{ \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Note that this means A is diagonalizable.

(b) Remember that T is supposed to be ‘reflection across a plane S ’. Explain what the eigenvalues and eigenvectors from (a) mean geometrically. What is their relationship to S ? Why does it make sense for the eigenvalues to be 1 and -1 ?

Solution. The $\lambda = 1$ eigenspace is a two-dimensional plane, consisting of vectors $\mathbf{v}$ that are unchanged by the linear transformation. In other words, this is the actual plane of reflection S . Vectors *in this plane* are not altered by the transformation.

The $\lambda = -1$ eigenspace is the *normal line* L , perpendicular to S . If we take a vector $\mathbf{v} \in L$ and reflect it across S , then we get precisely $-\mathbf{v}$ since the vector was exactly perpendicular to S .

The fact that A is diagonalizable is very helpful for understanding the transformation $T(\mathbf{x}) = A\mathbf{x}$. Since we have a basis of eigenvectors, it's possible to express any $\mathbf{x} \in \mathbb{R}^3$ as $\mathbf{x} = \mathbf{s} + \mathbf{n}$, for some $\mathbf{s} \in S$ (some linear combination of the two eigenvectors for $\lambda = 1$) and $\mathbf{n} \in L$ (a scalar multiple of the $\lambda = -1$ eigenvector). Then $T(\mathbf{x}) = \mathbf{s} - \mathbf{n}$, which means, in words, that “the S part of $\mathbf{x}$ is unchanged, but the L part gets negated”.

(c) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the “averaging transformation”:

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{3}(x_1 + x_2 + x_3) \\ \frac{1}{3}(x_1 + x_2 + x_3) \\ \frac{1}{3}(x_1 + x_2 + x_3) \end{bmatrix}.$$

Find all eigenvalues and eigenspaces for T . Explain your answer (what does it mean in terms of ‘averaging’?)

Solution. The algebra is similar to parts (a) and (b), though a little bit easier. You should find that the characteristic polynomial is $-\lambda^3 + \lambda^2 = -\lambda^2(\lambda - 1)$. So the eigenvalues are $\lambda = 1$ or 0 .

Here is a conceptual answer to guess the $\lambda = 1$ eigenvector directly: it should be a vector $\mathbf{v}$ such that applying T does not change the vector. In other words,

all three coordinates should *already* be averaged. So $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ works (clearly

$T(\mathbf{v}) = \mathbf{v}$ for this vector).

On the other hand, if $\mathbf{v}$ is a $\lambda = 0$ eigenvector, that means $T(\mathbf{v}) = \mathbf{0}$, so the coordinates of $\mathbf{v}$ should average out to 0. (Equivalently, the sum of the

coordinates should be 0). So we should guess vectors like $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and so on.

Since the multiplicity of $\lambda = 0$ was 2, we know that this eigenspace has dimension at most 2. So, as soon as we find two linearly independent eigenvectors, we know they must span the whole eigenspace. Therefore this

eigenspace is spanned by $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

(5) (Rotations)

- (a) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be rotation by $\pi/3$. Compute the characteristic polynomial of T , and find any eigenvalues and eigenvectors.

Solution. The characteristic polynomial will be $\lambda^2 - \lambda + 1$. It has no real roots. So there are no real eigenvalues!

This makes sense geometrically because no vector ever gets transformed to a multiple of itself.

(Comment: if you've heard of complex numbers like $i = \sqrt{-1}$, you might wonder if you can find eigenvectors and eigenvalues using i . The answer is yes, and this is extremely useful. But we're sticking to real numbers in Math 308, so for our purposes, T has no real eigenvalues.)

- (b) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a rotation in $\mathbb{R}^3$ by $\pi/3$ around some chosen axis L , a line through the origin in $\mathbb{R}^3$. **Without computing any matrices**, explain why $\lambda = 1$ is always an eigenvalue of T . What is the corresponding eigenspace?

Solution. If $\mathbf{v}$ is a vector in L , the axis of rotation, then $T(\mathbf{v}) = \mathbf{v}$. (If $\mathbf{v}$ is actually *on* L , then $\mathbf{v}$ 'swivels in place' but doesn't actually change.) Therefore $\mathbf{v}$ will be an eigenvector of eigenvalue 1. So, the $\lambda = 1$ eigenspace will be L itself.

- (6) Find a 3×3 matrix A with eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ with $\lambda = 1$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ with

$$\lambda = 2 \text{ and } \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ with } \lambda = 10.$$

(**Hint:** A must be diagonalizable, $A = PDP^{-1}$. Figure out P and D , then compute A directly.)

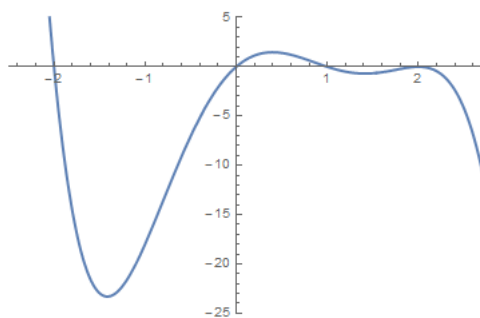
Solution. Since we're given the eigenvectors and eigenvalues, we can just set

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & & \\ & 2 & \\ & & 10 \end{bmatrix}$$

and set $A = PDP^{-1}$. After some computation, this works out to:

$$A = \begin{bmatrix} 16 & -3 & -3 \\ \frac{44}{3} & -\frac{4}{3} & -\frac{10}{3} \\ \frac{46}{3} & -\frac{11}{3} & -\frac{5}{3} \end{bmatrix}.$$

(7) Here is the graph of the characteristic polynomial of a 5×5 matrix A :



You may say “not enough information” for any of the following. Justify your answers.

(a) What polynomial has the graph shown above?

Solution. The roots of the polynomial are $-2, 0, 1, 2, 2$. Therefore the polynomial is $p(x) = -(x+2)x(x-1)(x-2)^2$. Note the double root at 2. The degree of the polynomial has to be 5 since A is 5×5 so it has to have 5 roots when counted with multiplicity. The negative sign in front comes from looking at the graph and noting that as $x \rightarrow -\infty$, $p(x)$ needs to be positive. Similarly, as $x \rightarrow \infty$, $p(x)$ needs to go to $-\infty$.

(b) What are the eigenvalues of A ?

Solution. The eigenvalues of A are the roots of its characteristic polynomial. Therefore, they are $-2, 0, 1, 2$.

(c) Compute $\text{nullity}(A + 2I)$.

Solution. Note that $\text{Null}(A + 2I)$ is the eigenspace of $\lambda = -2$. This eigenspace must be 1-dimensional since the dimension of the eigenspace cannot exceed the multiplicity of the eigenvalue and its dimension is at least one.

(d) What can you say about the value of $\text{rank}(A - 2I)$?

Solution. The nullity of $A - 2I$ is the dimension of the eigenspace of eigenvalue 2. We don't quite know the dimension of this eigenspace. It could be 1 or 2. Therefore, by the rank nullity theorem, $\text{rank}(A - 2I)$ is $5 - 2 = 3$ or $5 - 1 = 4$.

(e) Compute $\det(A - \lambda I)$.

Solution. This is the characteristic polynomial of A . This is the polynomial in (a) with $x = \lambda$. In other words it is the polynomial $-(\lambda + 2)\lambda(\lambda - 1)(\lambda - 2)^2$.

(f) What can you say about $\det(A + I)$?

Solution. If $\det(A + I) = 0$ then -1 is an eigenvalue of A which is not true. Therefore, $\det(A + I) \neq 0$. However we can estimate the value of $\det(A + I)$ by looking at the graph since $\det(A + I)$ is the value of the characteristic polynomial at $\lambda = -1$. From the graph this looks to be about -15 .

(g) Do the columns of A span $\mathbb{R}^5$?

Solution. Since 0 is an eigenvalue of A , we know that A has a kernel that is bigger than $\{0\}$. This means that the columns of A cannot span $\mathbb{R}^5$.

(h) Is A diagonalizable?

Solution. We don't know. It would have been if A had 5 distinct eigenvalues. If the eigenspace of 2 is 2-dimensional then A is diagonalizable.

(8) Suppose $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ with $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ is a linear transformation such that

- $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$ lie in the kernel of T , and
- all vectors of the form $(x_1, x_2, 0, 0)$ are reflected about the line $2x_1 - x_2 = 0$.

(a) Compute all the eigenvalues of A and a basis of each eigenspace.

Solution. Since A has a nonzero nullspace, 0 is an eigenvalue. The eigenspace of 0 is at least 2-dimensional since $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$ are linearly independent and their span is the x_3, x_4 plane in $\mathbb{R}^4$. Only points of the form $(*, *, 0, 0)$ can possibly be sent to something nonzero by T . Points on the line $2x_1 - x_2 = 0$ don't move under T . This line is spanned by $(1, 2, 0, 0)$. Therefore 1 is an eigenvalue with eigenvector $(1, 2, 0, 0)$. The eigenspace of 1 is the line $2x_1 - x_2 = 0$. Any point on the line perpendicular to the above line in the x_1, x_2 plane is reflected across the line $2x_1 - x_2 = 0$. Therefore, -1 is also an eigenvalue and its eigenspace is the span of $(2, -1, 0, 0)$. We conclude that A has eigenvalues, 0, 1, -1 with eigenspaces of dimension 2, 1, 1 respectively.

(b) Is A invertible? Explain.

Solution. A is not invertible since it has a nonzero nullspace.

(c) Is A diagonalizable? If yes, write down its diagonalization (you can leave it as a product of matrices). If no, why not?

Solution. This matrix has 3 eigenvalues 0, 1, -1 contributing four linearly independent eigenvectors: $\{(0, 0, 1, 0), (0, 0, 0, 1), (1, 2, 0, 0), (2, -1, 0, 0)\}$. Therefore A is diagonalizable, as

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & 1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}^{-1}.$$

(9) (a) If $f(t) = x_1 e^{-kt} \cos(\omega t) + x_2 e^{-kt} \sin(\omega t)$, then $f'(t)$ has the same form with coefficients y_1, y_2 .

- (i) Find the matrix A such that $y = Ax$.
- (ii) Find the characteristic polynomial of A .
- (iii) What can you say about eigenvalues of A ?
- (iv) Interpret your answer to (iii) as a calculus statement. That is, explain how your answer to (iii) could have been predicted from a basic fact of calculus.

- Solutions.** (i) $\begin{bmatrix} -k & \omega \\ -\omega & -k \end{bmatrix}$
(ii) $(\lambda + k)^2 + \omega^2$
(iii) no real eigenvalues
(iv) the only solution to $f'(t) = \lambda f(t)$ is $Ce^{\lambda t}$, which is not a function of the type considered.

Comment: Complex eigenvalues are $\lambda = -k \pm \omega i$, and $Ce^{(-k \pm \omega i)t}$ is of the type considered (with x_2 imaginary).

- (b) If $f(t) = x_1 \sin(t) + x_2 \cos(t) + x_3 t \sin(t) + x_4 t \cos(t)$, then $f'(t)$ has the same form with coefficients $y_1, \dots, y_4$. Same questions (a)-(d) as previously.

- Solutions.** (i) $\begin{bmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
(ii) $(\lambda^2 + 1)^2$
(iii)-(iv) same as in (a).